NEXUS ACADEMY - LECTURE NOTES

Limits & Differentiation Master Handout

Course: HSC Higher Mathematics (MTH-201) | Faculty: Tanvir Ahmed (Senior Faculty Lead)

Location: 965/1 Middle Monipur, Mirpur-2, Dhaka-1216

Key Topics Covered:

1. Fundamental Limits: sin(x)/x, ln(1+x)/x, and e^x limits
2. Derivatives of Trigonometric, Inverse and Exponential Functions
3. Chain Rule and Implicit Differentiation Techniques
4. Tangent, Normal, and Rate of Change Applications
5. Successive Differentiation and Leibniz Theorem

Official Study Material for Enrolled Students Only. Nexus ELMS Compliant.